Literature Review

The use of image processing in robotic arms, especially in applications such as goalkeeper robots, has been widely investigated. Many studies focus on achieving real-time object detection, accurate recognition, and efficient control of robotic movements. Early work by Karthikeyan et al. [1] explored an active tracking system that relied on hardware components such as laser diodes, steering mechanisms, and photoresistors to obtain two-dimensional coordinates. Instead of using complex image processing algorithms, their approach emphasized a simpler real-time tracking method that reduces computational load while still achieving reliable tracking performance.

Vision-based robotic systems have also been studied extensively. For example, Higa et al. [3] presented a system that uses a single web camera combined with image processing techniques to determine the position of objects for assistive robotic arms. Their work demonstrates how vision systems allow robotic arms to interact accurately with objects in their environment, which is particularly important for tasks like goalkeeping where precise and timely interception is required. In another study, Jia et al. [5] implemented color-based image processing methods to perform object sorting. Their approach highlights how color detection can be effectively used for real-time identification and control in robotic arm applications.

Image processing has also enabled more natural and intuitive ways of controlling robotic systems. Choudhary et al. [2] proposed a gesture-based control method where robotic arms respond to human hand gestures captured through vision systems. Similarly, Lengare et al. [4] used MATLAB to track human hand movements and translate them into commands for robotic arm motion. These approaches show how visual data can be used to create more interactive and user-friendly control mechanisms.

Another commonly used technique in robotic vision systems is color recognition. Szabo et al. [6] utilized color-based detection to control and position the SCORBOT-ER III robotic arm without requiring preprogrammed movements. This work demonstrates how visual feedback can provide flexible and adaptive control for robotic systems in both industrial and educational environments. Likewise, Dhayalini et al. [7] applied basic image processing techniques in a mobile robot platform designed for waste sorting, further showing how simple vision-based methods can be applied effectively across different robotic tasks.

More recent research has introduced advanced image processing techniques and machine learning algorithms. Kucarov et al. [8] developed a fully automated system that combines computer vision with image processing to accurately detect objects and determine their three-dimensional positions. Such capabilities are highly relevant for goalkeeper robots, where precise spatial awareness is necessary to intercept moving objects. In another example, Ilangovan et al. [9] applied deep learning techniques, particularly convolutional neural networks (CNNs), for weed detection in agricultural robots. Their work illustrates how modern machine learning approaches can significantly improve robotic perception and decision-making.

Overall, previous research shows that image processing plays a fundamental role in the development of robotic arm systems. Techniques range from simple color detection methods to advanced deep learning algorithms, all contributing to improved object detection, positioning accuracy, and system control. These capabilities are essential for dynamic applications such as robotic goalkeepers, where fast response and precise movement are critical for successful performance

Citations:
[1] https://www.paperdigest.org/paper/?paper_id=doi.org_10.1109_iccsp.2013.6577187
[2] https://www.paperdigest.org/paper/?paper_id=doi.org_10.1109_ichpca.2014.7045349
[3] https://www.paperdigest.org/paper/?paper_id=doi.org_10.14738_tmlai.24.309
[4] https://www.paperdigest.org/paper/?paper_id=doi.org_10.1109_pervasive.2015.7087039
[5] https://www.paperdigest.org/paper/?paper_id=doi.org_10.1109_eit.2017.8053385

[6] https://www.paperdigest.org/paper/?paper_id=doi.org_10.1109_eeeic.2017.7977643
[7] https://www.paperdigest.org/paper/?paper_id=doi.org_10.1109_icisc.2018.8398980
[8] https://www.paperdigest.org/paper/?paper_id=doi.org_10.1109_ines56734.2022.9922607
[9] https://www.paperdigest.org/paper/?paper_id=doi.org_10.1088_1742-6596_2923_1_012003

Proposed Plan for the Literature Review

To conduct the literature review effectively, the research will follow a structured plan that focuses on identifying, analyzing, and summarizing relevant studies related to image processing techniques used in robotic arm systems, particularly those applied in goalkeeper robots.

The first step will involve collecting and reviewing academic papers, journal articles, and conference publications that discuss image processing methods in robotics. Special attention will be given to techniques used for object detection, tracking, and real-time decision making, since these are essential for the operation of a robotic goalkeeper.

After gathering the relevant sources, the studies will be categorized based on the type of image processing techniques they use. These categories may include basic approaches such as color detection and object tracking, as well as more advanced methods like machine learning and deep learning algorithms. This classification will help in understanding the evolution of image processing techniques and their advantages in robotic applications.

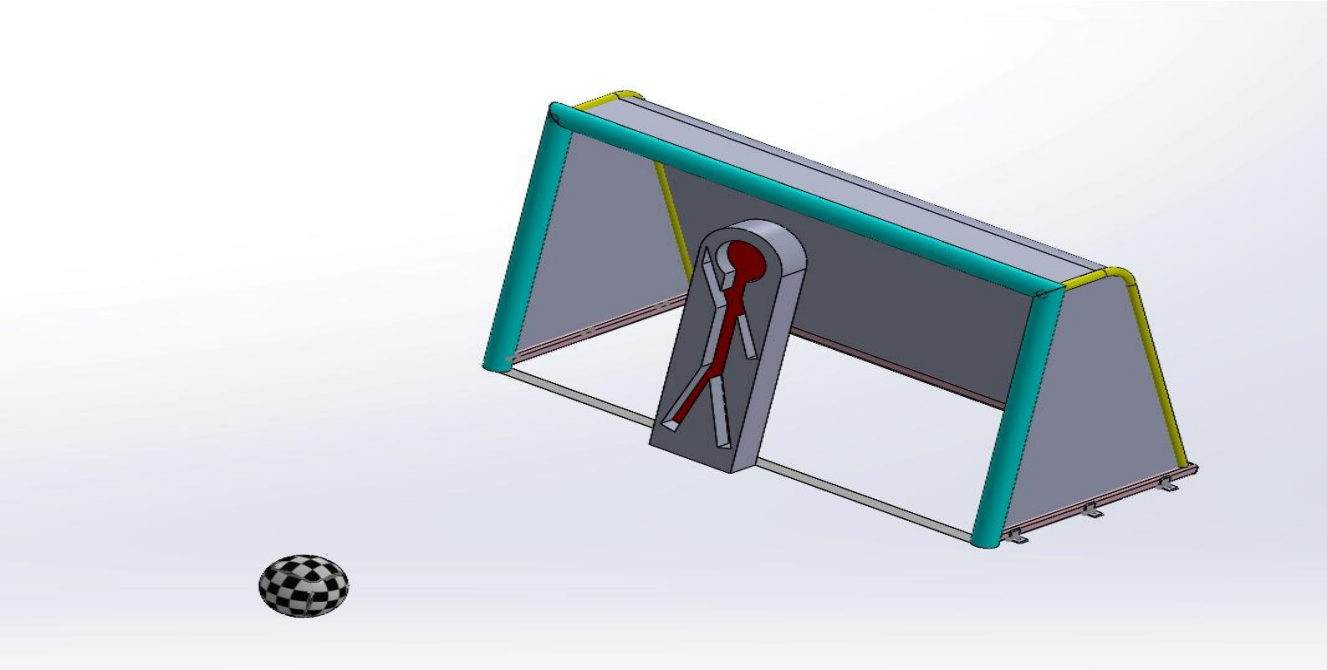
The next step will be analyzing the strengths and limitations of each approach, particularly in terms of accuracy, speed, computational requirements, and suitability for real-time robotic control. The review will also examine how different systems integrate image processing with hardware components such as cameras, sensors, and robotic arm mechanisms.

Finally, the findings from the reviewed literature will be summarized to identify the most suitable techniques for the proposed goalkeeper robotic arm project. Based on this analysis, a clear direction will be established for selecting the image processing method and system architecture that will be implemented in the project.

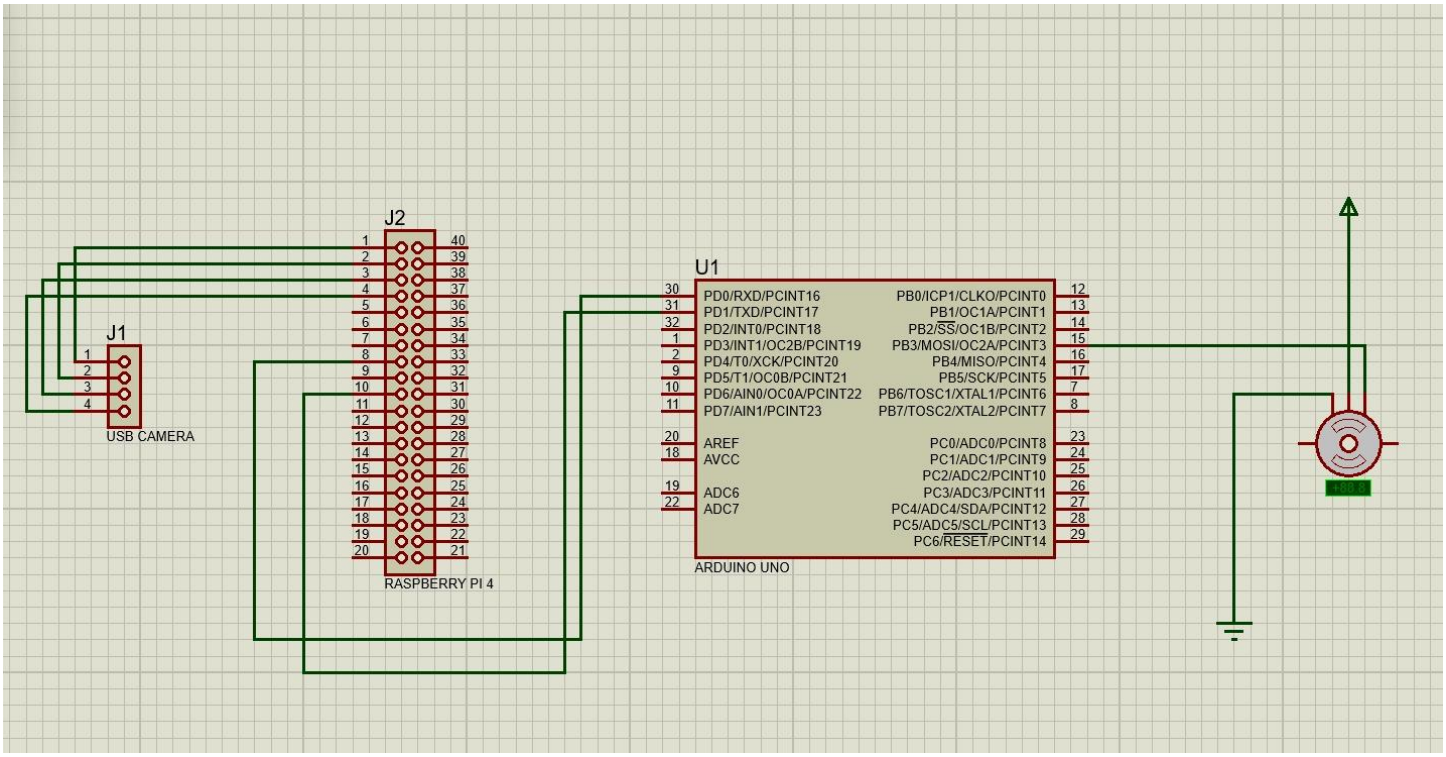
Component list

Component Category	Recommended Item	Purpose/Role +4
Main Processor	Raspberry Pi 4B or 5	Image processing (OpenCV/Python) and main decision-making.
Microcontroller	Arduino Uno or Nano	Receiving control signals from Pi and driving the motors.
Vision Sensor	Raspberry Pi Camera Module 3	High-speed image acquisition (connected via CSI ribbon).
Actuators	High-torque DC Servos (e.g., MG996R)	Moving the arm/paddle to block the ball.
Motor Driver	PCA9685 (if using multiple servos)	Relieving the Arduino from PWM generation overhead.
Power Supply	LiPo Battery (e.g., 2S 7.4V)	Powering the motors separately from the logic boards.
Structure	Aluminum/PLA/Acrylic	SolidWorks-designed chassis and arm links.

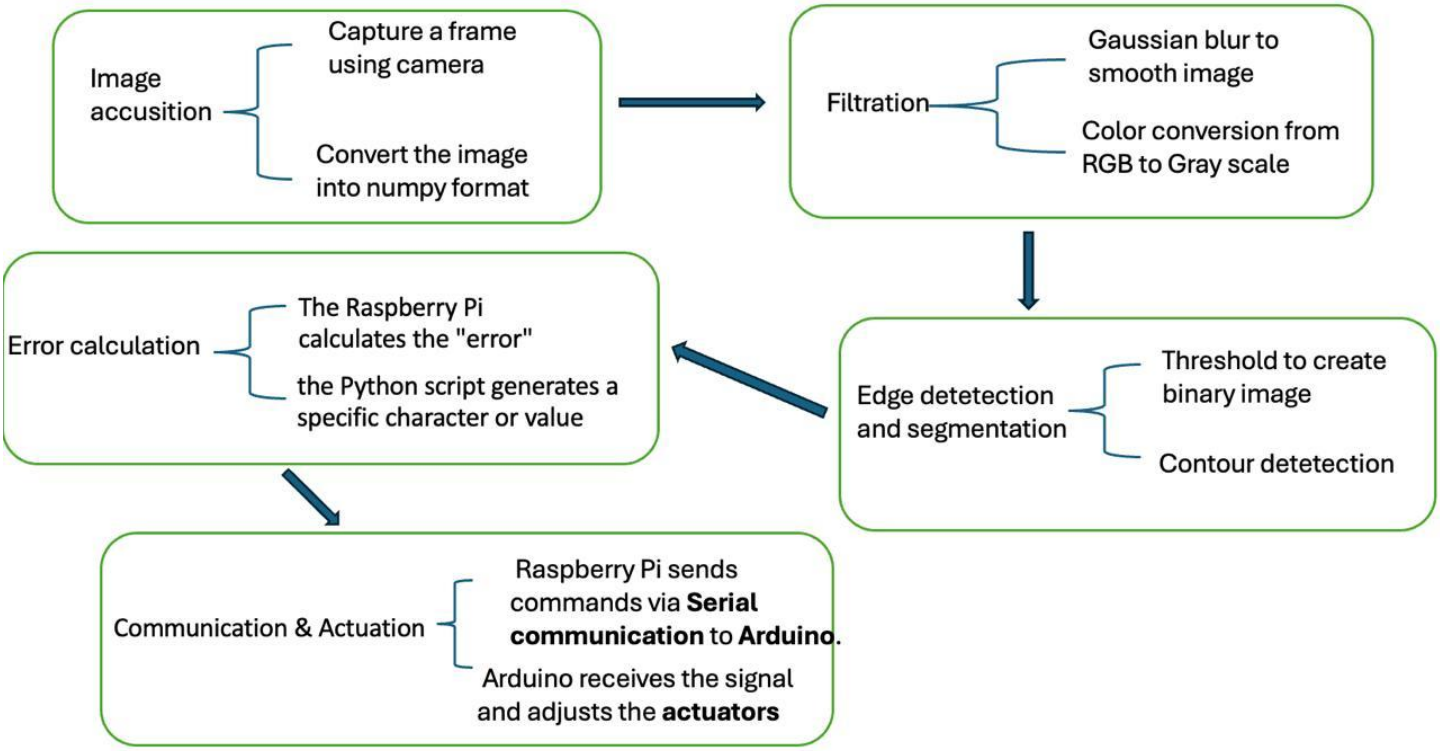
Solidworks



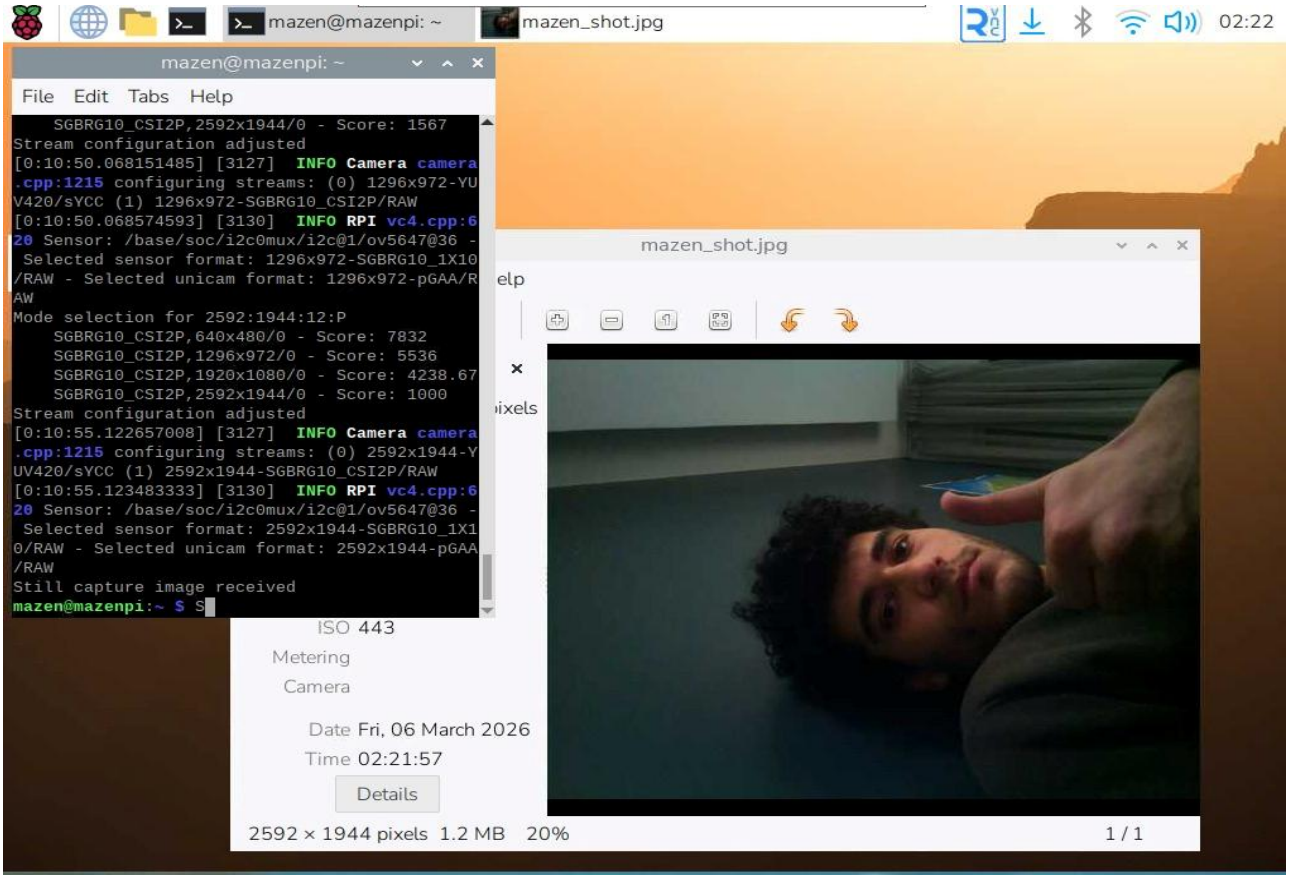
Circuit



Flowchart



Captured Image



Python Script

[2:34 AM, 3/6/2026] mazen emad: from picamera2 import Picamera2

```
# Initialize the camera without any arguments in the brackets
picam2 = Picamera2()

# Configure for a still image
config = picam2.create_still_configuration()
picam2.configure(config)

# IMPORTANT: We only call start() and capture_file()
# We skip picam2.start_preview() to avoid the VNC crash
picam2.start()
picam2.capture_file("/home/mazen/Desktop/mazen_final.jpg")
picam2.stop()

print("Photo saved to Desktop as mazen_final.jpg")
```

[2:34 AM, 3/6/2026] mazen emad: from picamera2 import Picamera2

```
picam2 = Picamera2()



config = picam2.create_still_configuration()
picam2.configure(config)



picam2.start()
picam2.capture_file("/home/mazen/Desktop/mazen_final.jpg")
picam2.stop()
```

```
print("Photo saved to Desktop as mazen_fina
```